

Wanda L. McCrae

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Professional Summary and Objective

Strong experience with backend development using Python and SQL. Experience with front-end Django and React frameworks, JavaScript, HTML, and CSS with an emphasis on accessibility. Excellent communication skills and ability to work in a collaborative team environment, including cross-departmental communication. Currently leading the backend design for the Tow Zone Alerts startup.

Key Career Highlights

- Leading the design and implementation of Tow Zone Alerts' backend architecture and account services API, fostering collaboration among team members to achieve a unified and effective solution using Docker, FastAPI/Python, and SQLAlchemy.
- Designed, implemented, and deployed Headline Editor to GitHub pages, allowing users to search the NY Times archives by year and month and edit the headline for the selected article.
- Implements CI/CD pipelines for quick project updates.
- Designed, implemented, and shipped production firmware as a senior hardware engineer for Telephonics Corporations.

Technical Skills

Programming: Python 3, JavaScript ES6+, SQL, HTML5, CSS, Agile Methodology, Docker

AI: Claude Code, DeepSeek

System Design: Microservices, REST APIs, Domain-driven design, Monoliths, Message Passing, SaaS

API Security: OAuth

Front-End: React, React Hooks, DOM Manipulation, Bootstrap, Tailwind CSS, Chakra UI

Back-End: Django 4, JSON, FastAPI, PostgreSQL, RabbitMQ, Twilio, JSON Encoders, SQLAlchemy

Programming Experience

FullStack Developer, Scam Job Reporter | scam-job-reporter.netlify.app 2026

- Built a full-stack anonymous reporting platform using FastAPI (Python) and React 18, backed by PostgreSQL, with JWT-based admin authentication, rate limiting, and security headers middleware deployed via Docker and Railway.
- Implemented comprehensive form validation and security including Google reCAPTCHA v2 integration, client-side profanity filtering, timing-safe credential verification (secrets.compare_digest), and CORS/security header configuration.
- Designed and maintained a RESTful API with paginated full-text search across multiple fields, endpoint-level rate limiting via SlowAPI, and automated CI/CD deployment to PythonAnywhere via GitHub Actions SSH pipeline.
- Developed accessibility-first frontend with dark mode and dyslexia-friendly font toggling (localStorage persisted), responsive mobile design, screen reader compatibility, and 41 React components using hash-based SPA routing.

FullStack Developer, Headline Editor | wlmccrae.github.io/headline-editor-ghpage 2024 - present

- Deployed to GitHub pages, allowing users to search the NY Times archives by year and month and edit the headline for the selected article.
- Designed and styled the responsive, accessible Single Page App using React, CSS, and Chakra UI.
- Deployed a FastAPI backend to use the NY Time archive API to receive articles for a given month based on user input.

Backend Web Developer, Tow Zone Alerts (Startup) | towzonealerts.com 2023 - present

- Leading the design of the backend architecture and account services API, fostering collaboration among team members to achieve a unified and effective solution.
- Collaboratively designing the FastAPI-based microservice with a PostgreSQL database used for user account creation, authentication, and maintenance.
- Utilizing GitHub and Docker containerization for development as part of a remote engineering team.
- Supporting React-based front end development as needed.

Front End Developer, Simple Weather | itsstheweather.netlify.app 2023 - present

- Deployed to Netlify, allowing internet users to get the local weather for any USA city.
- Designed and styled the Single Page App using React.js and CSS.
- Integrated 3rd party API calls to retrieve date/time, location, and weather information based on user input.

FullStack Developer, ShapeShifters | wmccrae.gitlab.io/shapeshifters 2023

- Designed and styled the fitness meetup single page app using React with Redux Toolkit and Tailwind CSS.
- Utilized FastAPI to query a PostgreSQL database and call our RESTful API routes.
- Implemented a PostgreSQL database to allow authenticated users to create, read, update, and delete event information.

Senior Hardware Engineer, Telephonics (Farmingdale, NY)

- Designed, implemented, and shipped production firmware, which was deployed in air traffic control and military aircraft radar systems.
- Coded and ran unit tests to exercise production code.
- Participated regularly in code reviews.

Frontend Web Developer, Lotusland Fine Art (Volunteer) | lotuslandfineart.com

- Developed and managed a multi-page fine art photography website using HTML and CSS.
- Enhanced site engagement by weekly blog post updates with meticulous coding, images, and CSS styling.
- Successfully transitioned the website to a self-hosted WordPress installation.

Other Experience

Software Engineering Immersive Resident, Galvanize (New York, NY) - Remote 2023

- Provide one-on-one and small group technical and project management support to students.
- Conduct code reviews on students' project work.
- Be present during class time to answer questions or provide support to instructors.
- Share observations on student progress with the Cohort Leads and instructors.

UX Designer, Creative Luv (New York, NY) - Remote 2022

- Reviewed customer's website in Firefox, Chrome, and Safari on desktop and phone to determine the site's responsiveness in different environments.
- Researched and discussed with customer options for porting their site to a modern WYSIWYG editor so they can update prices and product details as needed.

Education

Syracuse University, M.S. Computer Engineering

New Jersey Institute of Technology, B.S. Computer Engineering

Hack Reactor, Advanced Software Engineering Certificate